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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GLY 417 — Past Questions & Materials
by Renance

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Paper 1 of 1 — GLY 417 sequence stratigraphy

Level 400 • First Semester • Handout

APPLICATION OF SEQUENCE STRATIGRAPHY TO HYDROCARBON EXPLORATION AND EXPLOITATION

In the quest for hydrocarbons, both the exploration and exploitation programs can be greatly enhanced by applying seismic sequence stratigraphic analysis. This technique provides the explorationist with the capability to recognize, discover and evaluate new hydrocarbon reservoirs and to reduce the risk in management's decision making. The play concepts, source, seal and traps associated with different reservoir sands and their associated lowstand, transgressive and highstand systems tracts are an important aspect of this type of analysis.

Seismic Sequence Stratigraphy Analysis is the integration of high resolution biostratigraphic and paleobathymetric data, well-log signatures and seismic-reflection profiles. This methodology permits the geologist and geophysicist to divide an outcrop, well- and stratigraphic section and seismic section into a series of third-order depositional sequences, chronostratigraphic maximum flooding surfaces and sequence boundaries. The maximum flooding surface condensed section, maximum flooding surface and sequence boundaries are age dated on the basis of high resolution biostratigraphy and stratigraphic analysis is to succeed. These are correlated with the global cycle chart. Subsequently, these third order depositional sequences are subdivided into systems tracts on the basis of characteristic lithology, Paleobathymetry, well-log signatures and seismic reflection patterns.

The most important contribution to seismic sequence stratigraphy in the past years has been the addition and integration of high resolution biostratigraphy and paleobathymetry. The high resolution, biostratigraphy and paleobathymetry are an absolute must if seismic sequence data sets are extremely important because they provide a means of recognizing maximum flooding surface condensed sections which in turn provides an accurate means of identifying the maximum flooding surfaces on well-logs and seismic profiles which in turn provides a means of age dating these surfaces and correlating them with the global cycle chart.

The paleobathymetry provides a means of accurately identifying the water depth at which the reservoir sands were deposited which in turn provides a means to help select the various types of systems tracts that are associated with these shallow or deep water reservoir sands. For instance, the blocky well-log signature of both the deep water lowstand systems tract basin floor fan complex and the very shallow water prograding or transgressive systems tract incised valley fill can easily be differentiated based on the paleobathymetry. The maximum flooding surface condensed sections are also important because they are usually associated with source rocks and provide an excellent seal for hydrocarbons.

Sequence stratigraphy is the study of sedimentary rock relationships within a chronostratigraphic framework. The discipline is now mature, having evolved over a 50-year period that began with the concept and applications of seismic stratigraphy presented in a landmark publication by Payton (1977). Sequence stratigraphy provides a powerful tool for predicting the spatial and temporal distribution of reservoir, source, and seal rocks within a stratigraphic interval, at both

the exploration and the reservoir scales.

Sequence stratigraphic intervals are deposited in cycles of relative sea-level change (highstand to lowstand and vice versa) over a variety of time scales, from hundreds of millions of years to

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thousands of years. Superimposition of these cycles in time and space provides predictable stratigraphic patterns from terrestrial to deep-marine environments.

Various criteria have been recognized to identify these stratigraphic patterns and the contained facies. Once a single significant chronostratigraphic surface has been identified and mapped (on seismic lines, in well logs, or in outcrops), then the surface's sequence stratigraphic framework, and the facies contained above and below that surface, can be predicted. Reservoir, hydrocarbon source, and seal rocks also can be predicted in time (geologic and seismic) and space.

High-frequency parasequences- which are composed of laterally continuous shales, and of sandstones that have laterally and vertically variable reservoir quality and continuity- can have a major effect on reservoir performance. Thus, they should be identified as early as possible in the life of a field by sequence stratigraphic analysis.

In summary, sequence stratigraphy allows the rational and realistic predictions of the following during petroleum exploration and exploitation:

- a). Source Rock: the source rock is defined as an organically-rich sedimentary rock that has potential to generate or had generated petroleum. Examples of some common source rocks include shale, and limestone. Most of the shale and limestone associated with the transgressive systems tract (TST) and partly those of the lowstand systems tract (LST) have proven to be source rocks in most sedimentary basins worldwide.
- b). Reservoir rock: any porous and permeable rock can serve as a reservoir rock where petroleum may readily accumulate petroleum in commercial quantity. Examples of reservoir rocks are sandstone, limestone; any faulted, fractured, or weathered rock. Thick sandstone of the lowstand systems tract (LST) and the sandstone of the highstand systems tract (HST) have served as reservoir rocks.
- c). Seal or Cap rock: any compacted or non-pervious rock directly overlying a reservoir rock that can prevent further migration of petroleum from the reservoir rock e.g. shale. The shale overlying the thick sandstone of the lowstand systems tract (LST) acts as a seal or cap rock for petroleum that accumulates in the sandstone reservoir.
- d). Petroleum Trap: any geological feature that bars migration of petroleum away from the reservoir rock. Traps can be generally classified into three (3) namely structural traps (e.g. fold, and fault traps), stratigraphic traps (e.g., unconformity trap, sand pinch-out traps), and combination traps (e.g., diapiric structures). Sequence stratigraphic tools have played important roles in the exploration of petroleum traps especially the subtle stratigraphic traps.
- e). Sequence stratigraphy has also aided the regional correlations of the above petroleum elements.

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UNCONFORMITY

To the subsurface geologist it is apparent that unconformities which punctuate the stratigraphic column in outcrop become progressively less distinct as they are traced into subsurface sedimentary basins. Obviously, while erosion is proceeding on land, deposition continues in the sea.

Changes in lithologic character and variations in faunal content become less distinct basinward, with the result that in the basin centers unconformities lose their identity. Similar phenomena occur in outcrop, as is well known. A pronounced angular unconformity that is present in one place on the surface may disappear completely within a few miles.

There are basically two types of geological boundaries in the rock record namely: Bedding plane and unconformity surface. A bedding plane is a surface that separates a rock layer (bed) on top from another at the base. Bedding plane often indicates a short pause in deposition. An unconformity surface also separates two beds from each other; however, it often suggests a longer pause in deposition.

Unconformity may represent missing stratum or strata. Such missing strata may be due erosion that removes part of a stratum or strata before much younger stratum is deposited on the erosive surface, this erosive surface is termed as unconformity surface.

Unconformity may also evolve through non-depositional processes. Non-deposition may cause as a result of a prolong period of no deposition, or a period when certain tectonic process such a deformation may occur. When deformation is active, there will not be deposition in a basin where the deformation is operating. This will also lead to missing time in the rock record.

Some unconformities are evidenced of both non-deposition and erosion.

Unconformity often depicts missing time referred to as HIATUS. This missing time represents the age (or the time) of the missing stratum or strata.

Classification of unconformity

Numerous attempts have been made to classify unconformities; the most durable of these schemes classified them into four (4): (1) Nonconformity (2) Angular unconformity, (3) Disconformity, and (4) Paraconformity.

Nonconformity describes the erosion surface between plutonic igneous rocks and metamorphic rocks, below, and stratified sedimentary rocks, above. For example, the Cretaceous sedimentary rocks in Nigeria unconformably overly the Precambrian crystalline basement rocks (i.e., the plutonic igneous and /or metamorphic rocks). The time interval between the Precambrian and the Cretaceous when there was no sediment deposition or rocks in the rock record is the missing time (Hiatus). Secondly, oftentimes there may not be erosive surface.

Angular unconformity designates that variety which separates strata of quite different structure (Figure 1).

Disconformity has strata above and below essentially parallel, yet shows some evidence of erosion between the two sets of strata.

Paraconformity refers to the situation in which no apparent erosional break exists, but a time hiatus is nevertheless present (i.e. key fossil zones are missing).

Figure: An angular unconformity